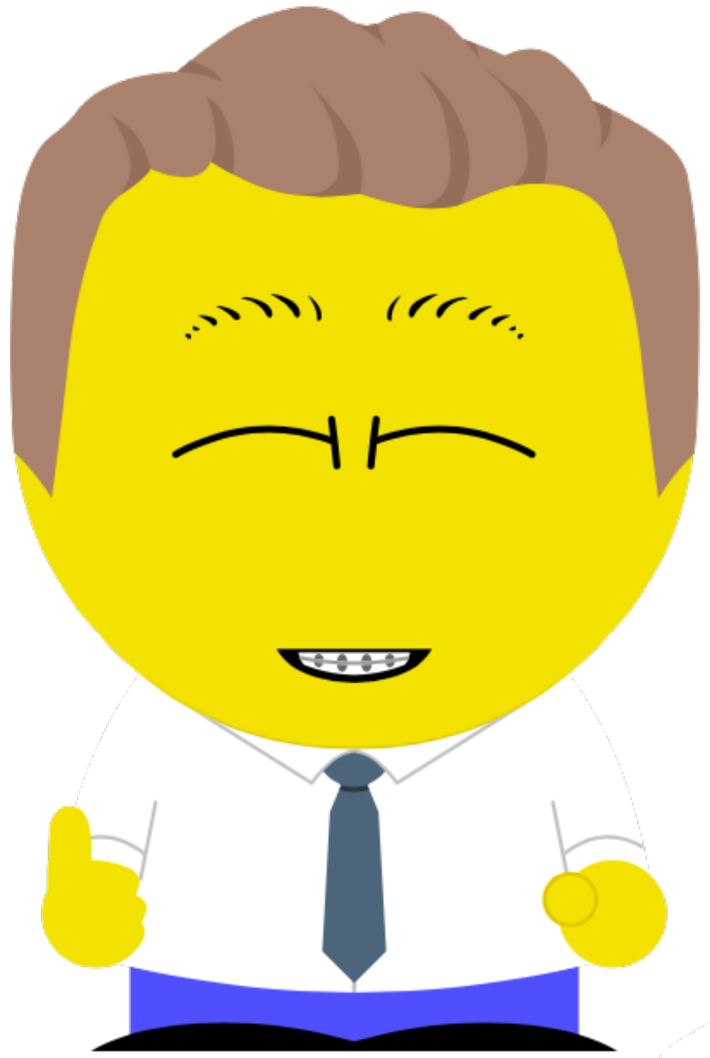
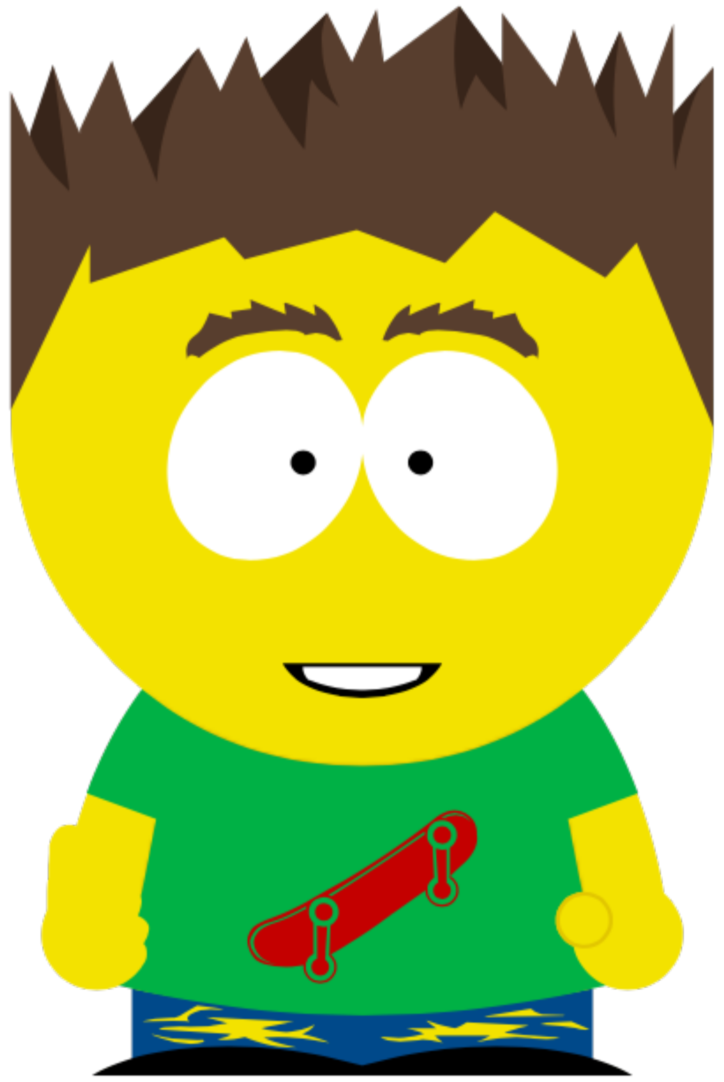
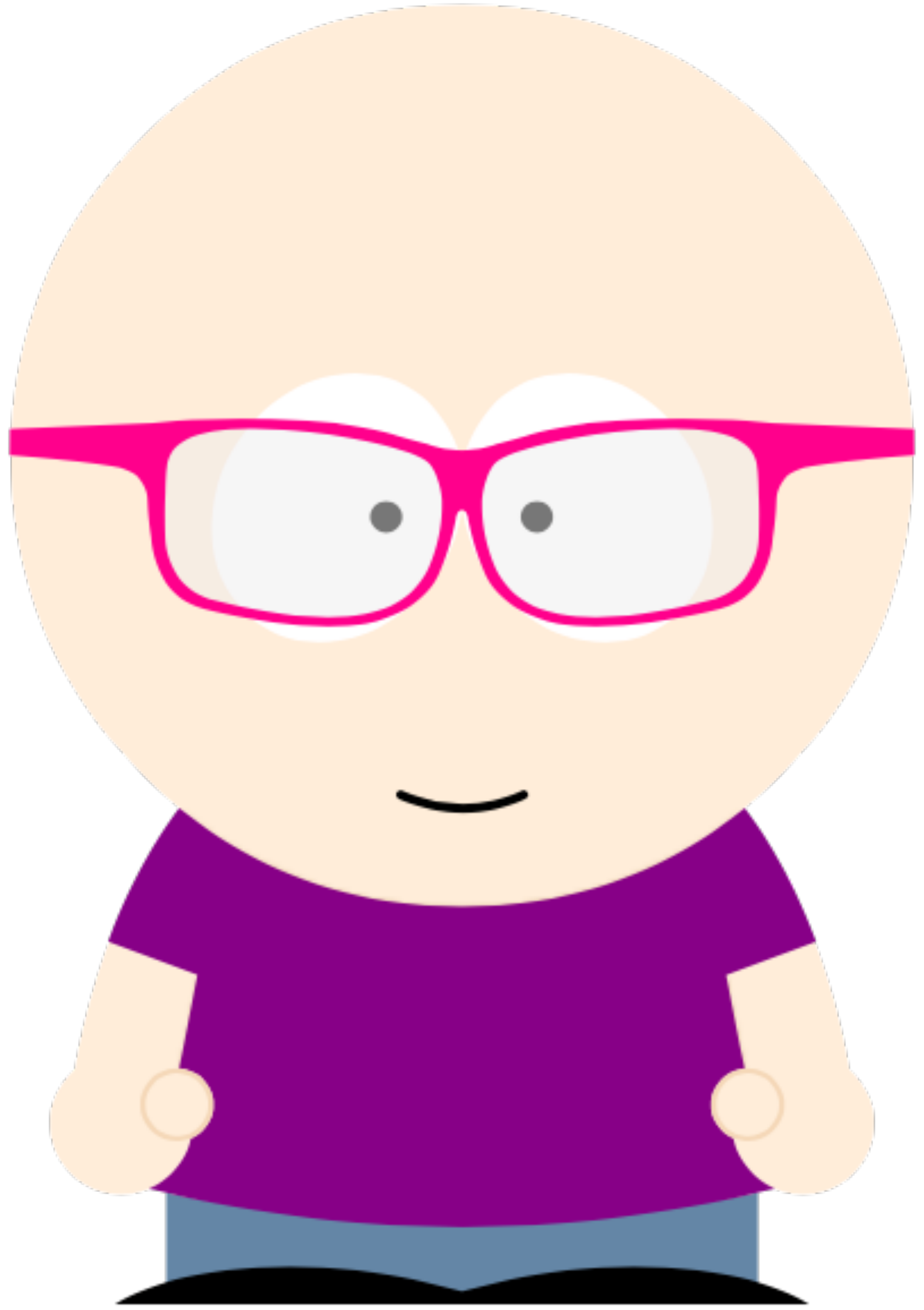






CSc 453
Compilers and Systems Software


Spring 2026

TuTh 12:30-1:45

Version: Jan 2, 2026

Course Description

- This course covers the design and implementation of translator-oriented systems software, especially compilers. Topics covered include lexical analysis, parsing, and syntax-directed code generation.
- This class has a significant programming component.

You are responsible for reading and understanding this syllabus. If you have any concerns or issues about the information in this document you should bring them up during the first week of class.

Instructor and Contact Information

- Christian Collberg
- GS 758
- 621-6612
- collberg@cs.arizona.edu
- Office hours: Open Door Policy
- https://ligerlabs.org
- https://collberg.cs.arizona.edu

Teaching Assistants

- Brixen, Bennett, bennettbrixen@arizona.edu
- Upadhyay, Utkarsh, upadhyay2@arizona.edu

Course Format and Teaching Methods

- Students are expected to prepare prior to class by
  - watching videos,
  - reading papers
  - reading sections from the text book,
  - etc
- Lectures will consist of discussions, exercises, and group activities.

Course Objectives

1. Introduction (2 lectures)
2. Lexical analysis (3 lectures)
3. Grammars + syntactic analysis (5 lectures)
4. Semantic analysis (4 lectures)
5. Miderm + exam reviews (3 lectures)
6. Intermediate Code (2 lectures)
7. Interpretation (1 lecture)
8. Code Generation (2 lectures)
9. Procedures (2 lectures)
10. Optimization (2 lectures)
11. Garbage collection (2 lectures)
12. Object oriented languages (2 lectures)

Topics

lexing

The Chomsky hierarchy, regular expressions, DFAs,
scanner implementation.

parsing

Context-free grammars, BNF, parse trees,
abstract syntax trees, Recursive Descent parsing.

semantic analysis

Attribute grammars, environments,
type-checking.

intermediate representations

stacks, tuples, trees,
intermediate-code generation from abstract syntax trees.

Topics

code generation

control-flow graphs,
code generation for arithmetic expressions, data-structure access, control-flow, and procedure calls.

code optimization

survey of techniques, peepholes optimization.

object-orientation

compiling Java-like languages, run-time class templates, inheritance.

garbage collection

reference counting, pointer-maps, mark-and-sweep.

Expected Learning Outcomes

We will...

- learn how compilers are constructed,
- learn how programming languages are designed,
- learn how to compile procedural languages,
- learn how interpreters work,
- learn how garbage collectors work,
- learn how object oriented programming is implemented.

Course Communications

Official course communication will be conducted through D2L.

Required Texts or Readings

1. Douglas Thain, *Introduction to Compilers and Language Design*, https://dthain.github.io/books/compiler/

2. T. Mogensen, *Introduction to Compiler Design*. https://link.springer.com/book/10.1007/978-0-85729-829-4

Required or Special Materials

A laptop to bring to class for in-class exercises.

Assignments and Examinations

The final grade will be calculated based on

- 8 take-home graded assignments.
- 1 midterm exam.
- 1 final exam.
- Class participation

Assignments

- Assignments 2-8 will be connected.
- That means that assignment *i* will depend on assignment *i-1*.
- If you have bugs in your assignment *i-1* you are responsible to fix them prior to attempting assignment *i*.
- If you don't, it is likely that those bugs will affect the correctness (and hence your grade) for assignment *i*.
- 7 assignments will be in Java, 1 in the gcc/clang version of C.

Final Examination

- The final exam is scheduled for **Wednesday, May 13, 2026,1:00pm-3:00pm.**

- Final Exam Regulations and Final Exam Schedule: https://registrar.arizona.edu/faculty-staff-resources/room-class-scheduling/schedule-classes/final-exams

Grading Scale and Policies

1. 1 Midterm: 26%
2. 1 Final: 26%
3. Attendance and class participation: 24%
4. 8 Take-home assignments: 8*3% = 24%

Grading Scale and Policies…

- Take-home assignments are due midnight.
  - Assignments handed in < 24 hours late get a 10% penalty.
  - Assignments handed in > 24 hours late will get a grade of 0.

- Grades will be calculated as follows:
  - 90-100: A; 80-89: B; 70-79: C; 60-69: D; 0-59: E

Attendance

- Each class you attend will count for 1% of your grade.
- Once you’ve attended 24 classes you’ve achieved the maximum.
- If you are not actively participating in class I may deduct points.
- If you don't attend the entire class hour I may deduct points.
- The midterm does not give you an attendance point.

Dispute of Grade Policy

Students must dispute a grade no later than 7 days after the grade has been handed back to them.

Tentative Scheduled Topic and Activities

- Below is a very preliminary schedule which may be adjusted based on instructor discretion.
- Videos and other online resources will be provided for most lectures. **They must be watched/read before class.**

#	Date			Lecture	Assignments
1)	Thursday	January	15	compilers-1	
2)	Tuesday	January	20	tiny-1	[Assignment out 1: tiny]
3)	Thursday	January	22	lex-1	
4)	Tuesday	January	27	lex-2	
5)	Thursday	January	29	lex-3	[Assignment due 1: tiny; Assignment out 2: lexing]
6)	Tuesday	February	3	grammars-1	
7)	Thursday	February	5	parse-1	
8)	Tuesday	February	10	parse-2	
9)	Thursday	February	12	parse-3	[Assignment due 2: lexing; Assignment out 3: parsing]
10)	Tuesday	February	17	parse-4	
11)	Thursday	February	19	sem-1	
12)	Tuesday	February	24	sem-2	[Assignment due 3: parsing; Assignment out 4: ast]
13)	Thursday	February	26	sem-3	
14)	Tuesday	March	3	REVIEW	
15)	Thursday	March	5	MIDTERM	
	Tuesday	March	10	SPRING BREAK	
	Thursday	March	12	SPRING BREAK	

#	Date	Lecture	Assignments
16)	Tuesday March 17	sem-4	
17)	Thursday March 19	interm-1	[Assignment due 4: ast; Assignment out 5: semantics]
18)	Tuesday March 24	interm-2	
19)	Thursday March 26	interp-1,javavm-1	
20)	Tuesday March 31	codegen-1	[Assignment due 5: semantics; Assignment out 6: interpretation]
21)	Thursday April 2	codegen-2	
22)	Tuesday April 7	proc-1	
23)	Thursday April 9	proc-2	[Assignment due 6: interpretation; Assignment out 7: procedures]
24)	Tuesday April 14	opt-1	
25)	Thursday April 16	opt-2	
26)	Tuesday April 21	gc-1	[Assignment due 7: procedures; Assignment out 8: codegen]
27)	Thursday April 23	gc-2	
28)	Tuesday April 28	oo-1	
29)	Thursday April 30	oo-2	[Assignment due 8: codegen]
30)	Tuesday May 5	REVIEW	

Classroom Behavior Policy

- You are encouraged to bring laptops to class. They will be used to work exercises in small groups.
- You should refrain from using these or other devices for other, non-class-related, purposes.
- If you who insist on performing non-class related task in class (including, but not limited to: working on other classes, chatting, surfing, socializing, using their phones, etc.) you will be asked to leave the classroom.

Incomplete (I) or Withdrawal (W)

- Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at https://catalog.arizona.edu/policy/courses-credit/grading/grading-system

Safety on Campus and in the Classroom

- For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): https://cirt.arizona.edu/case-emergency/overview

- Also watch the video available at: https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/app/me/ledetail;spf-url=common%2Flearningeventdetail%2Fcrtfy00000000000384

University-wide Policies link

- Links to the following UA policies are provided here: https://catalog.arizona.edu/syllabus-policies
- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy
- Class Recordings
- Additional Resources
- Preferred Names and Pronouns

Department-wide Syllabus Policies and Resources link

- Links to the following departmental syllabus policies and resources are provided here: https://www.cs.arizona.edu/cs-course-syllabus-policies
- Department Code of Conduct
- Illnesses and Emergencies
- Obtaining Help
- Confidentiality of Student Records
- Land Acknowledgement Statement

Subject to Change Statement

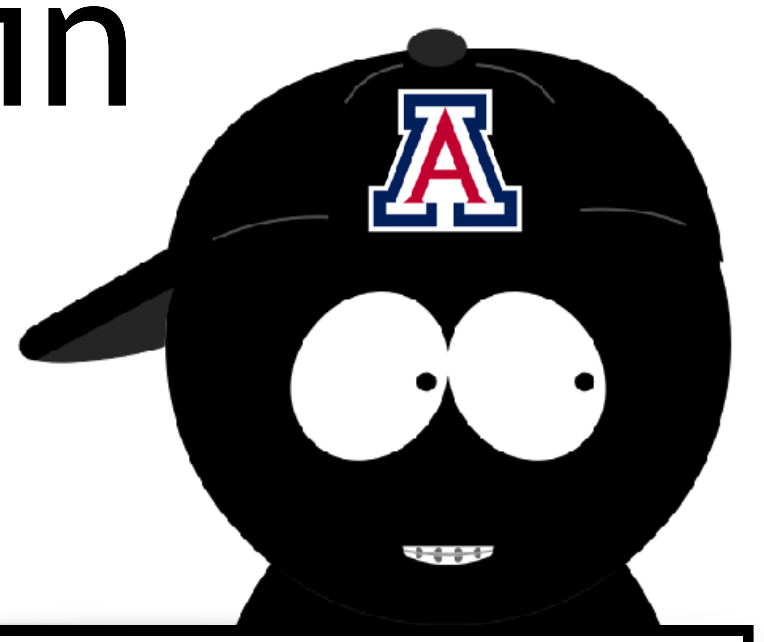
Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.

Use of AI

- All programming assignments in this class should be done
  - *by yourself*,
  - *without the use of AI or similar resources*, and
  - *without help from other humans*.
- It is a violation of academic integrity to *receive* help from other students, but also to *provide* help.
- If you have problems with the programming assignments you should seek help from the instructor and TAs.

Violation of the Spirit of Assignments

- Some years ago, we had a student who turned in a compiler that looked like this:



- This one weird trick worked because the instructor shared the test cases with the class.
- While, technically, the student didn't violate any written rules, they did violate the spirit of the assignment.

```
void my_compiler() {
   if (test_case_1)
      puts(output_1);
   else if (test_case_2)
      puts(output_2);
   else if (test_case_3)
      puts(output_3);
   ... ... ...
}
```

- Obviously, we will not accept any such behavior.